RCS Day 4

Supply, DWV, P-trap assembly, dry verification, and system coordination

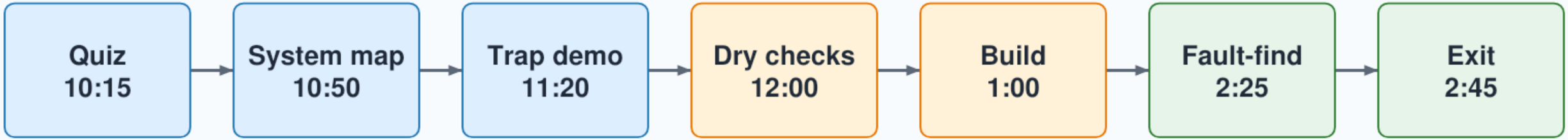
ALIGN FIRST. TIGHTEN SECOND.

Dry classroom verification only unless the instructor controls a separate test.

Today the goal is not to force fittings together. The goal is an aligned, explainable path.

Today's flow

Quiz -> model -> guided practice -> capstone lab -> QC



Morning: separate supply from drain-waste-vent functions.
Afternoon: dry-fit, inspect, fault-find, correct, and map a capstone path.
End: parts identified, fault documented, station dry and reset.

Today's targets

What successful work must demonstrate

Supply pressure

Shutoff isolation

Fixture use point

P-trap seal

DWV gravity path

Differentiate supply and drain-waste-vent functions
Explain fixture, trap, and shutoff roles
Plan a training plumbing path through framing
Assemble and troubleshoot a dry rolling-sink/P-trap module

The day's path

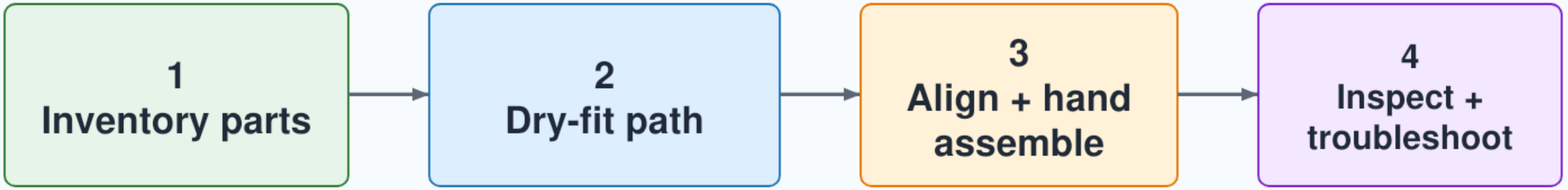
Trace the work before building it



Trace pressure toward the fixture and gravity away from it. Keep the full path accessible for inspection.

Controlled sequence

Release each stage only after its check



Orient washers before tightening. Correct cross-threading or damaged parts instead of applying more force.

Working roles

6 students; every result receives another set of eyes

Builder
assembles path

Inspector
checks parts + alignment

Documenter
maps + records fault

Rotate after the first dry assembly.
The Inspector names each part and checks thread start, washer orientation, and access.
Change one condition at a time.

Verification checkpoints

Do not trust appearance alone

Correct parts

Washer orientation

Threads start clean

Path aligned + accessible

Inspect before tightening.
A reliable diagnosis records expected condition, observed condition, one correction, and a dry recheck.

Safety boundaries

The lab begins only inside these controls

No uncontrolled water

Electrical zone separate

No unapproved drilling

Stop damaged parts

Wipe unexpected moisture immediately. Keep cords and electrical equipment out of the plumbing station. No structural penetration without approval.

Professional judgment

Safe work pauses when the evidence conflicts

Good path
dry-fit -> align -> tighten -> inspect

Bad habit
force threads -> hide leak path

Cross-threading and misaligned washers do not improve with confidence. Stop, disassemble, identify the fault, and correct it.

Quality criteria

Good physical work leaves visible proof

Inventory every fitting and washer.
Trace supply and drain paths separately.
Start slip nuts by hand.
Keep joints aligned and accessible.
Correct one fault, dry-reverify, and document.

Part does not seat? Stop forcing it.

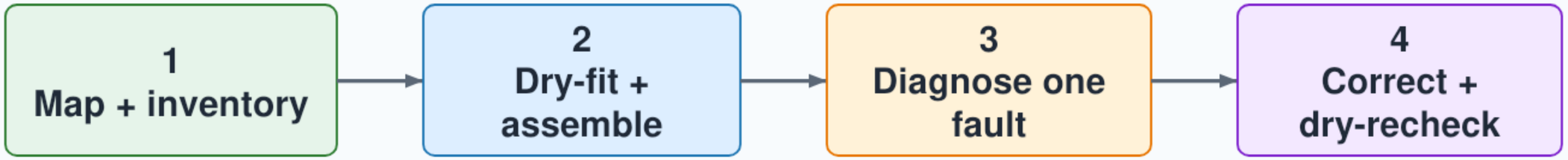
Lab release checklist

The station earns work only after the checks

Module inventory complete.
Supply and drain paths traced.
Parts inspected for cracks or damage.
Washers oriented before assembly.
No water/pressure test without authorization.
Capstone path avoids electrical boxes.

Afternoon lab

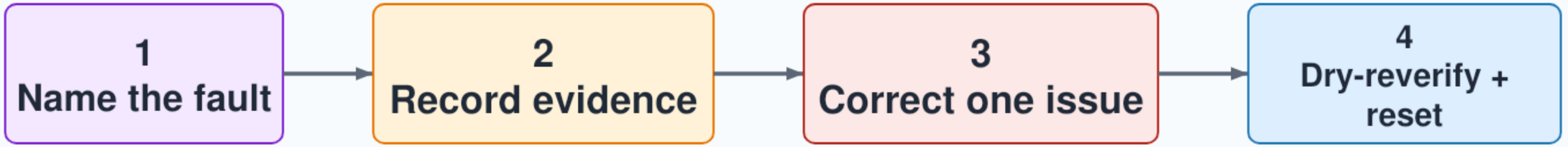
Slow sequence. Clear callouts. No surprises.



Rebuild the rolling sink/P-trap module, then map a coordinated plumbing pathway on the capstone wall.

Closeout order

End verified, documented, and reset



Keep the station dry. Record the part, condition, correction, and result before returning the module.

Evidence log

What students write down

System map	supply + drain	Done: _____
Parts	correct + sound	Pass: _____
Washers	orientation	Pass: _____
Threads	clean start	Pass: _____
Alignment	accessible path	Pass: _____
Capstone	coordinated route	Sketch: _____

No dry inspection, no approval. Alignment is easiest to correct before tightening.

Exit ticket

Three checks before dismissal

1. Draw the fixture-to-drain path through the P-trap.

2. Name one fault and the evidence that revealed it.

3. State today's water and drilling boundaries.

Module dry. Parts counted. Fault documented. Station reset.